

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — ACTIVE LAYERS 5-8 (RTL TO VERIFICATION)

Feedback, Verification & Ground-Truth Trust

1. Execution Signals

- Simulator Trace Logs
- Verilator Assertion Hits
- STA Negative Slack
- OpenROAD DRC Errors

2. Failure Triage

- Automated Bug Triage
- Root-Cause Classification
- Error Attribution Map
- Diagnostic Summary

3. Ground-Truth Checks

- Cycle-Exact Co-Simulation
- Physical STA Signoff
- Power / Area Floorplan Check
- Invariant Compliance

4. Trust Verdict

- Authoritative Verification
- Non-Delegable Human Gate

Verdict Gate:

ADVANCE / REJECT

Diagnostic Feedback & Requalification Loop